1.3 Technological Foundations

1.3.1 Early developments in aerial systems

The evolution of drone technology is rooted in early experimentation with unmanned flight that began in the early 20th century. Initial developments were driven by military interests seeking to perform reconnaissance and training missions without risking human pilots. Early systems such as the “Aerial Target” (1916) and the “Queen Bee” (1935) laid the groundwork for remotely piloted aircraft through basic radio control mechanisms and gyroscopic stabilization. The Cold War era saw significant advances in unmanned systems, particularly in surveillance and target drones, as nations recognized the strategic importance of persistent aerial observation. These formative years established the structural, aerodynamic, and control principles that continue to underpin modern UAV designs (Blom, 2010).Table 1.1 presents the chronological evolution of early aerial systems, tracing the progression from simple observation balloons and kites to the first radio-controlled aircraft. This table highlights the pioneering innovations that established the technological foundations of unmanned aerial vehicles (UAVs), emphasizing the gradual development of propulsion, remote control, and automation that shaped the future of drone technology.